POWERING POTENTIAL

ANALYTICS WITH POWER BI

Analytics with Power BI

- Clustering
- Time series forecasting

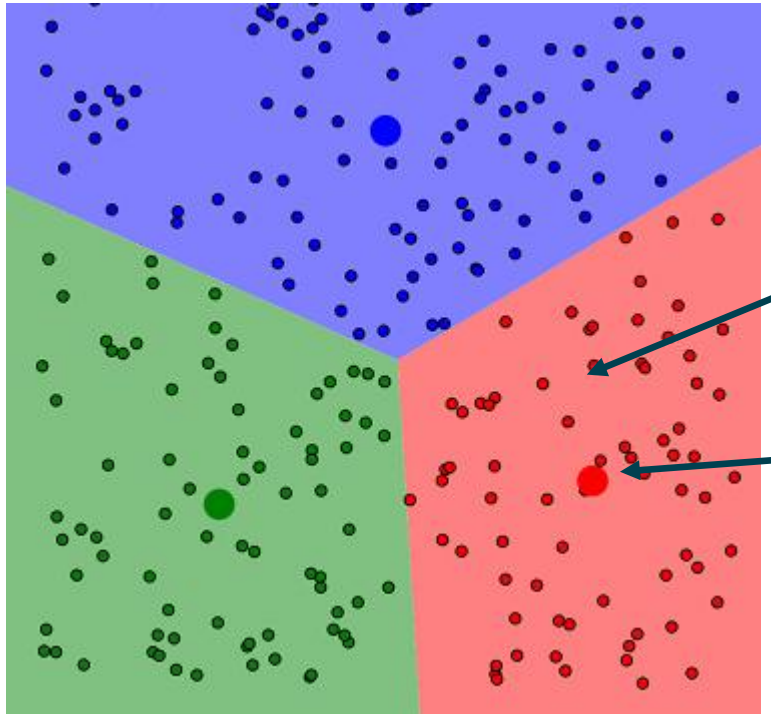
CLUSTERING

Clustering (Cluster Analysis)

- Analytical technique to group objects according to their **attributes**, so that objects belonging to the same group are much more similar to objects in the **same** group than to objects in other groups.
- Finds existing patterns in data
- **Unsupervised learning**
- Once the groups are determined and labelled, the clustered data can be used to perform **classification**.
- One of the most popular clustering method is **k-means clustering**. K is the number of clusters (groups) that is input to the method.

CLUSTERING

https://www.naftaliharris.com/blog/visualizing-k-means-clustering/



cluster

centroid

K-means clustering is based on:

- Distances (this means only quantitative attributes can be used)
- Centroids

CLUSTERING

- Each object (data point) is shown as a dot which represents its attributes.
- A centroid is the **average representative** of a group. It is a calculated data point, where the value of each attribute is found by taking the average of the attribute's values from the members of the group.

How k-means clustering works:

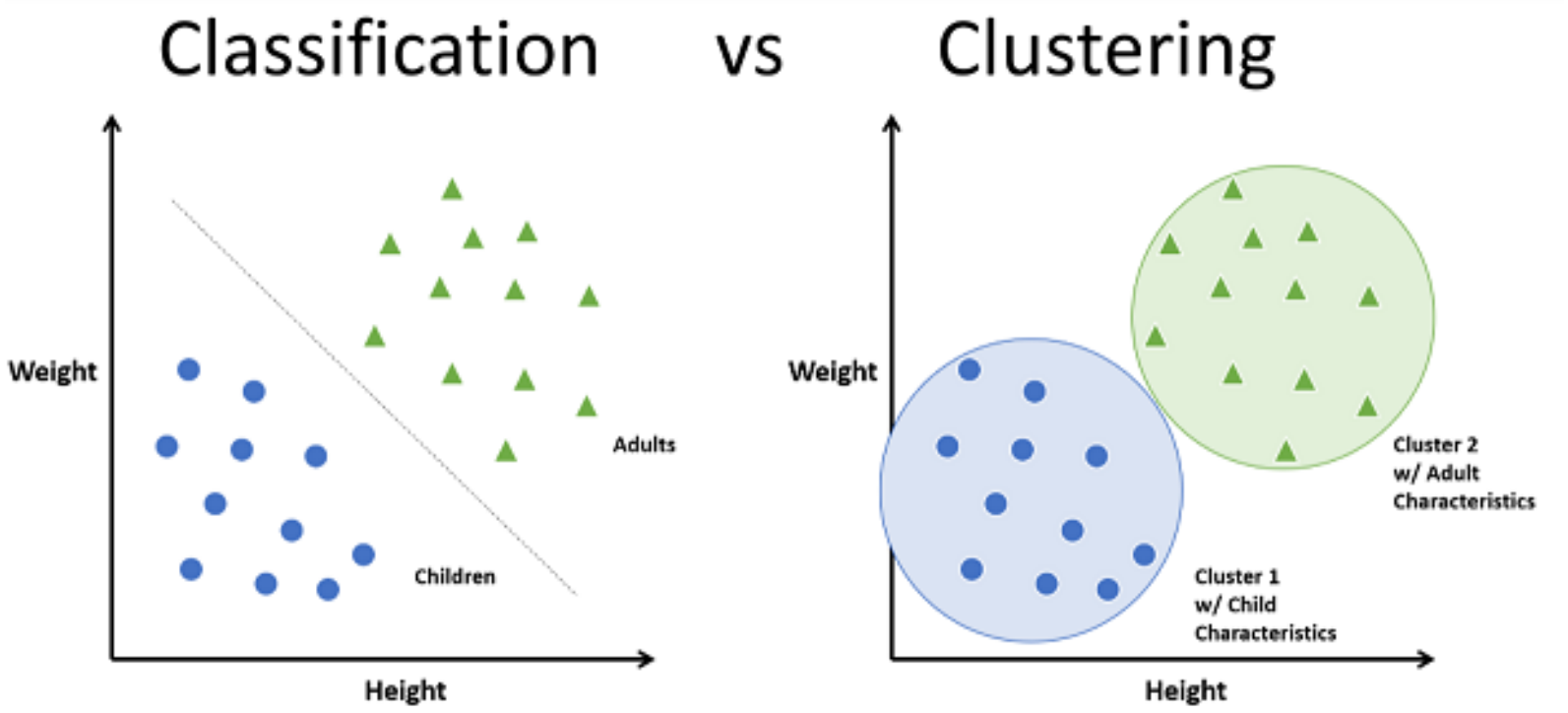
1. Select k random points for initial centroids.
2. Assign each data point to its nearest centroid.
3. Re-calculate the centroid for each group (average of the points in the group).
4. Repeat (2) and (3) until the centroids and the groups stabilise.
5. The best number of clusters is found by running the algorithm for different values of k and selecting the case with the minimum and majority compact clusters.

➢ **Try a very good interactive animation of k-means clustering:**

https://www.naftaliharris.com/blog/visualizing-k-means-clustering/

CLUSTERING VS CLASSIFICATION

Clustering determines and labels the groups. This data can then be used to perform **classification** – determining to which labelled group new data points belong.



Source:

https://www.analyticsvidhya.com/blog/2021/05/what-why-and-how-of-spectral-clustering/

CLUSTERING IN POWER BI

In Power BI, clustering can be invoked from a:

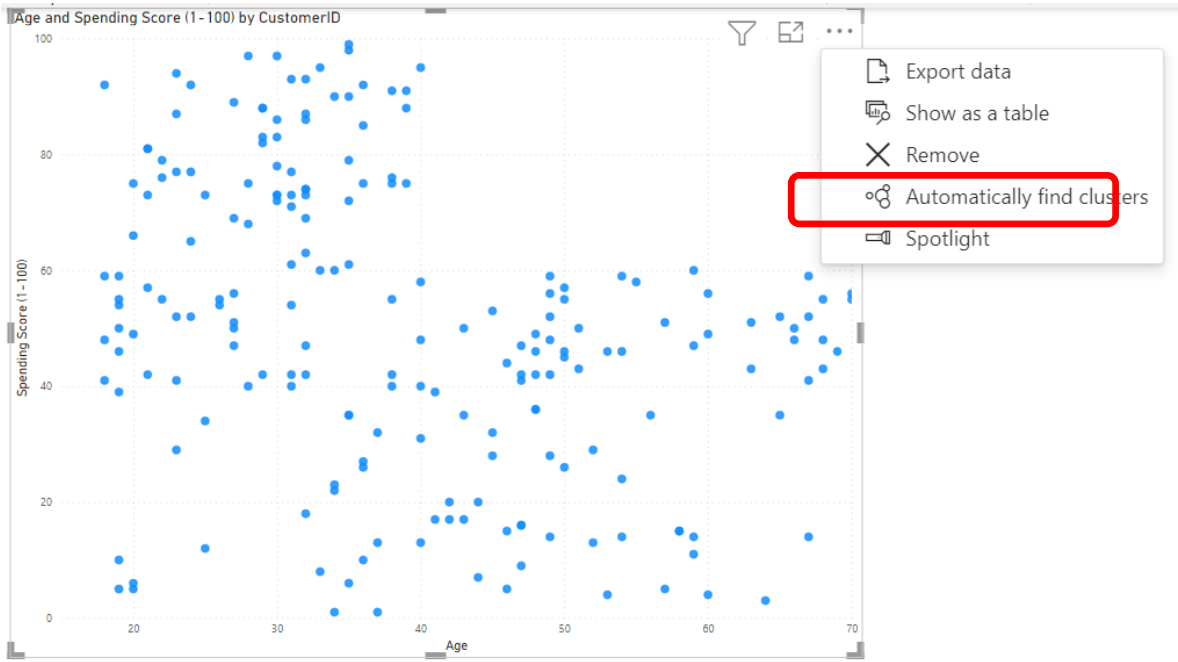
- **scatter chart** – for up to two attributes.
- **table** – more than two attributes.

Let's try to determine what types of customers a company has.

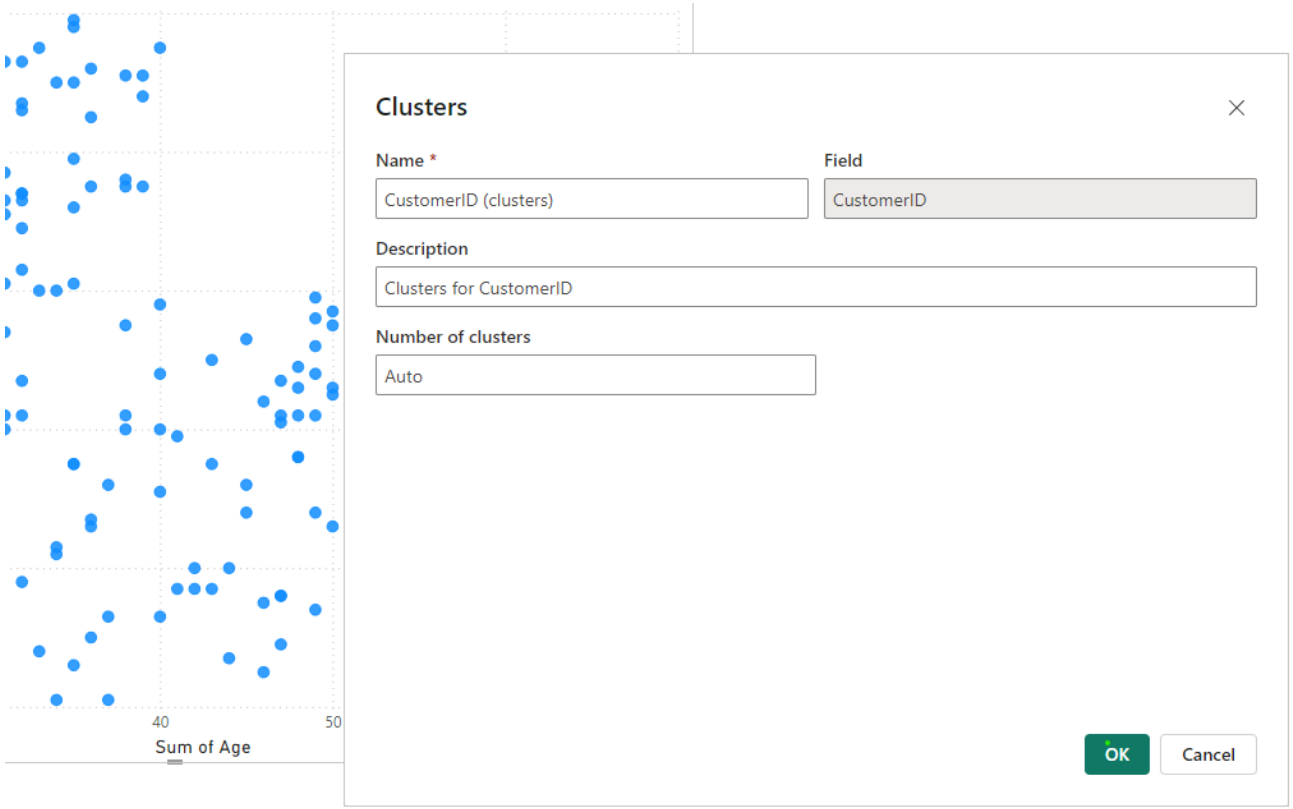
- We will use the data in file **Segment_Customers.csv**.
- The numeric attributes there are: **Age**, **Annual Income (000)**, and **Spending Score (1-100)**.
- We will use **Age** and **Spending Score (1-100)** with a scatter chart and all attributes with a table.

CLUSTERING IN POWER BI: SCATTER CHART

- Load file **Segment_Customers.csv**
- Select **scatter chart**:
  - X axis: Age
  - Y axis: Spending Score (1-100)
  - Values: CustomerID
- Select **Automatically Find Clusters** from the menu on the top right.

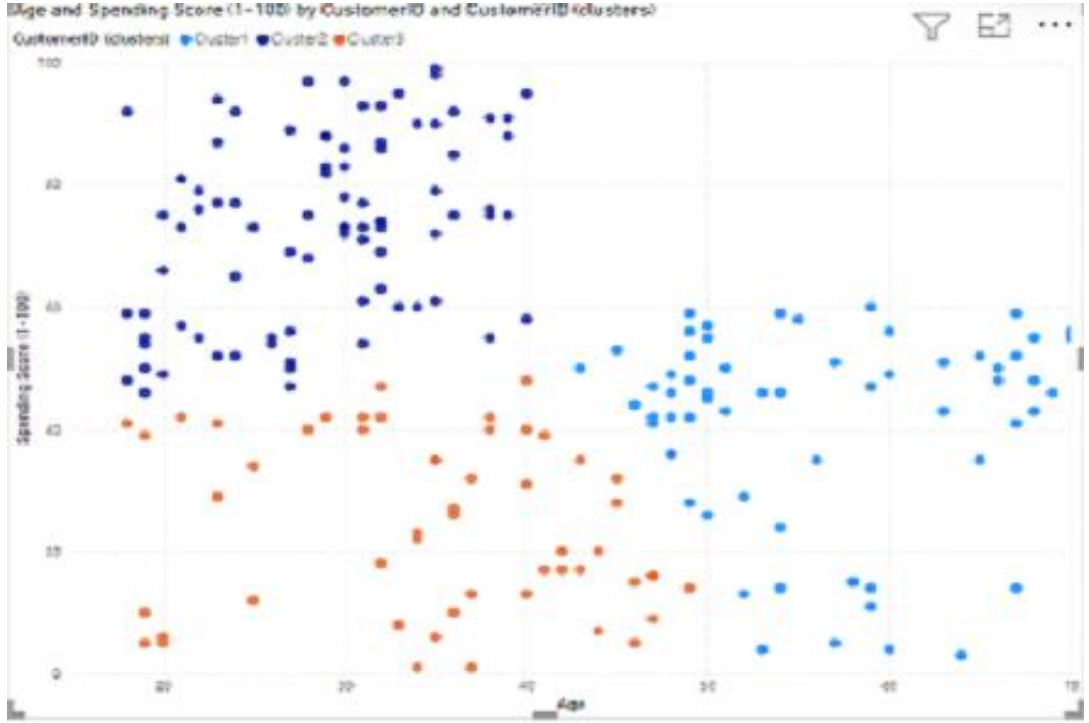


CHOOSING CLUSTERS



CLUSTERING IN POWER BI: SCATTER CHART

- You can change the name and the description.
- You can input the number of clusters or leave it at **Auto** for Power BI to determine the best number.

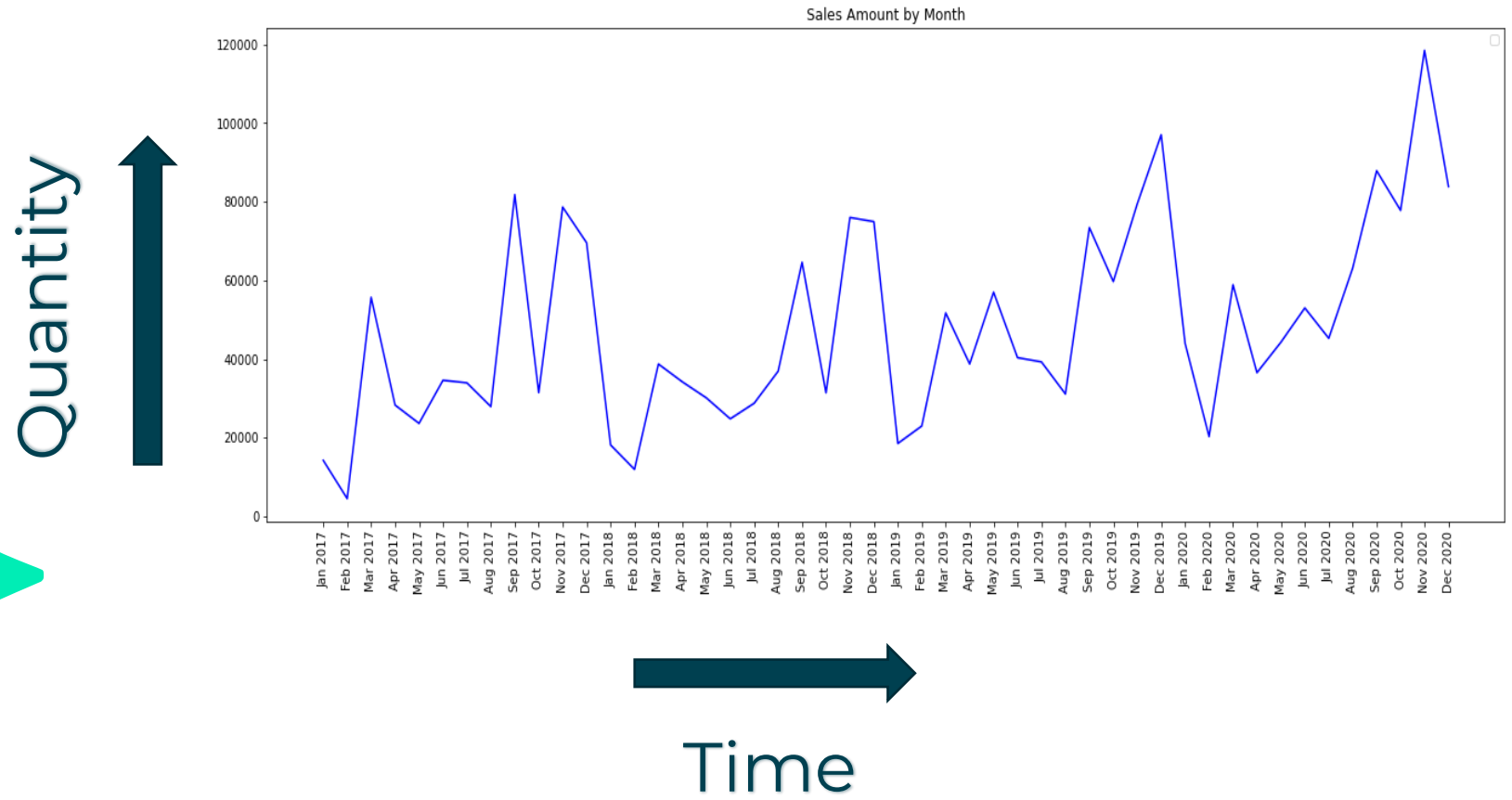


Three clusters were automatically identified.

- How would you label (describe) them?

TIME SERIES

A time series measures quantity over time. The interval between consecutive observation has to be constant (e.g., hour, day, week).



TIME SERIES

Time series forecasting is based on the fact that you can predict future values by analysing the past values. It relies on **repeating patterns**.

What affects the quality of results:

- Forecast horizon (how far ahead we forecast)
- Clarity of the pattern
- Presence of strong factors affecting the pattern (causal factors)

TIME SERIES

The pattern of the data in a time series has several components:

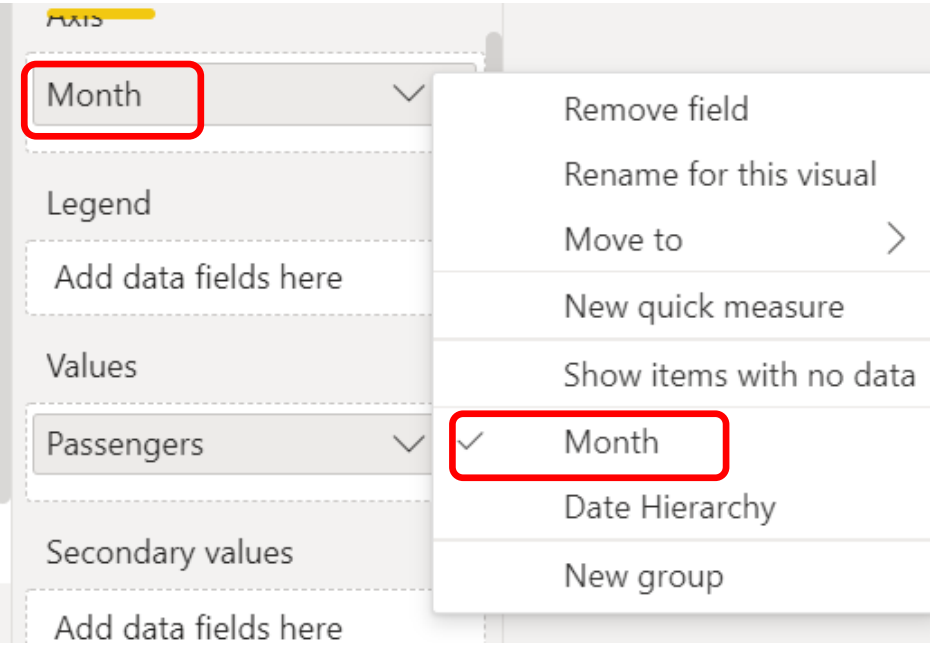
- **Trend:** overall long-term direction of the series (up or down):
  → House prices
- **Seasonal:** repeating pattern over a fixed period (the length of the period does not matter as long as it stays constant):
  → Demand for ice cream over a year
- **Cyclical:** repeating pattern over a period that is not fixed (usually long term):
  → Economy cycles – expansion, contraction
- **Random (remainder, noise, irregular):** what is left after removing all other components

TIME SERIES FORECASTING IN POWER BI

In Power BI, **time series forecasting** is an analytical function used with the **line chart** visual.

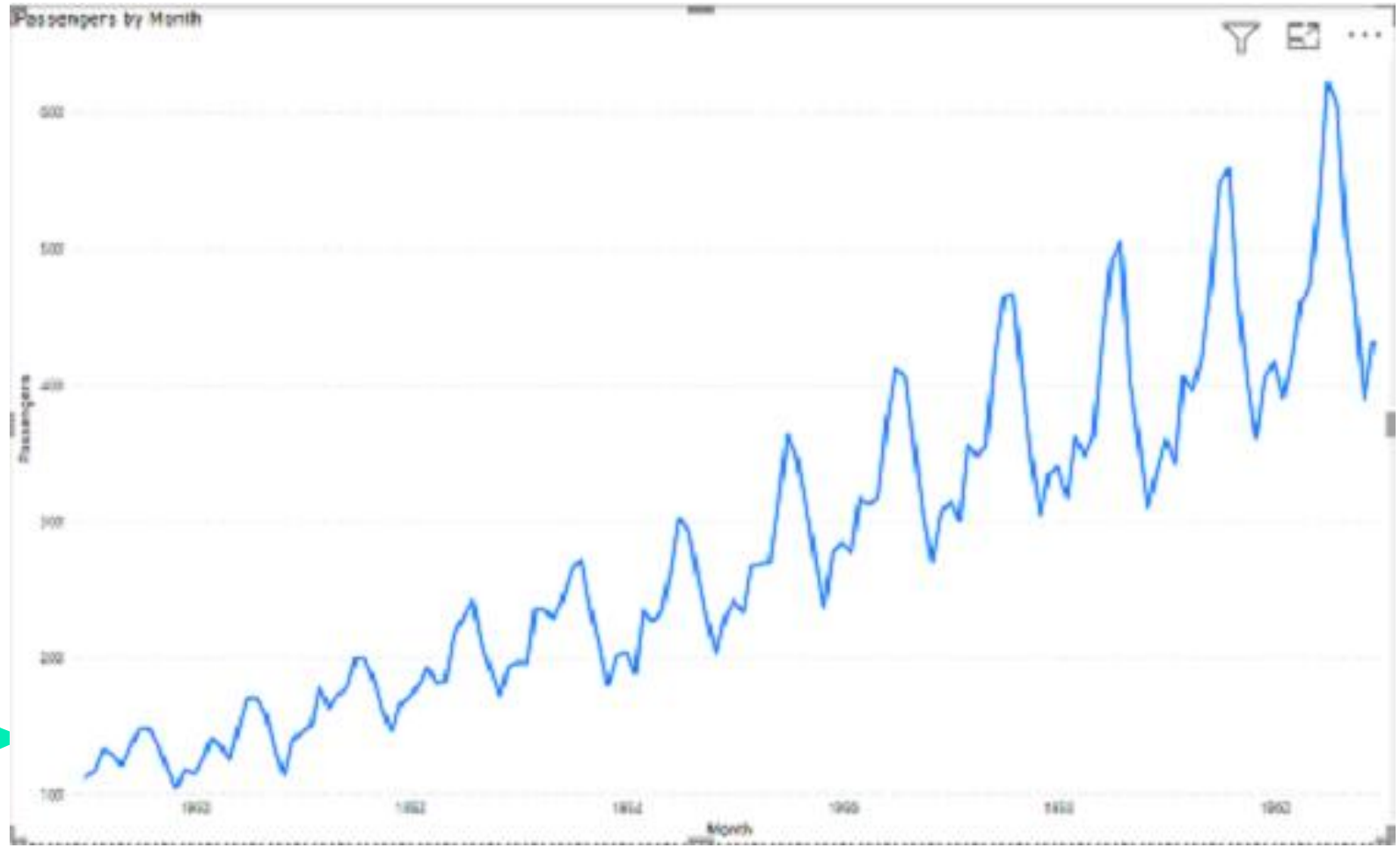
We will use data for the monthly number of passengers of a USA air company in the period 1949-1961.

- Load the data from the file **AirPassengers.csv**.
- Select the **line chart** visual:
  - X Axis: Month
  - Y Axis: Passengers
- For the month, select **Date**, not Date Hierarchy:



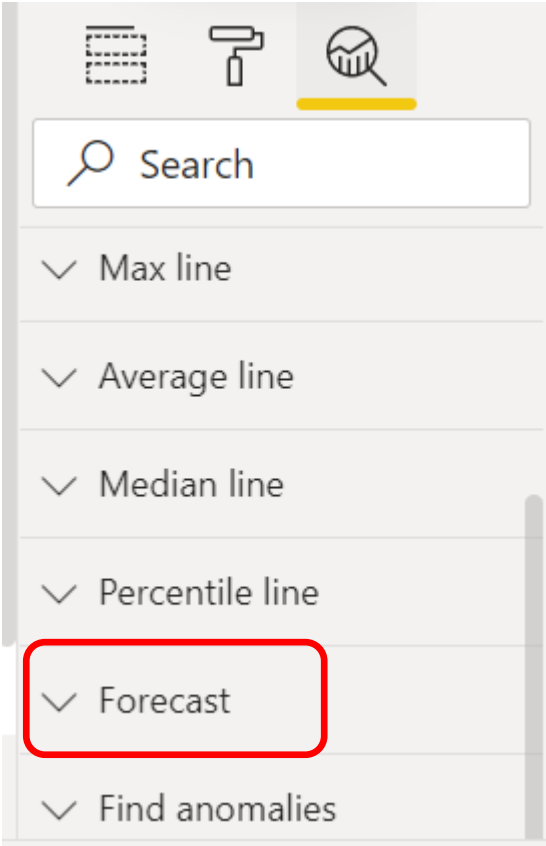
TIME SERIES FORECASTING IN POWER BI

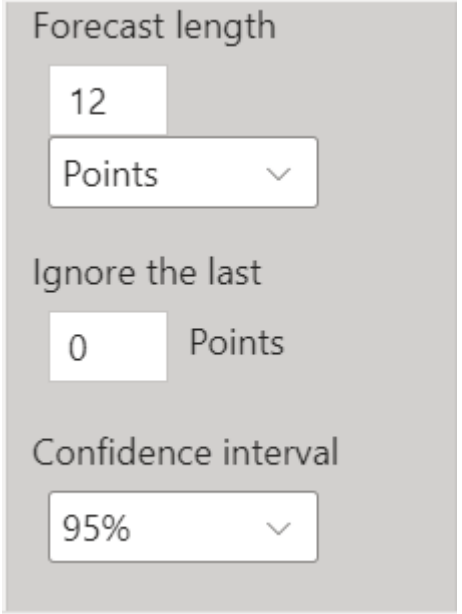
This is the history:



TIME SERIES FORECASTING IN POWER BI

- To forecast, select the **Analytics** tab (the magnifying glass). It also lists a number of statistics that can be computed and plotted to compare with each value.
- Select **Forecast**.

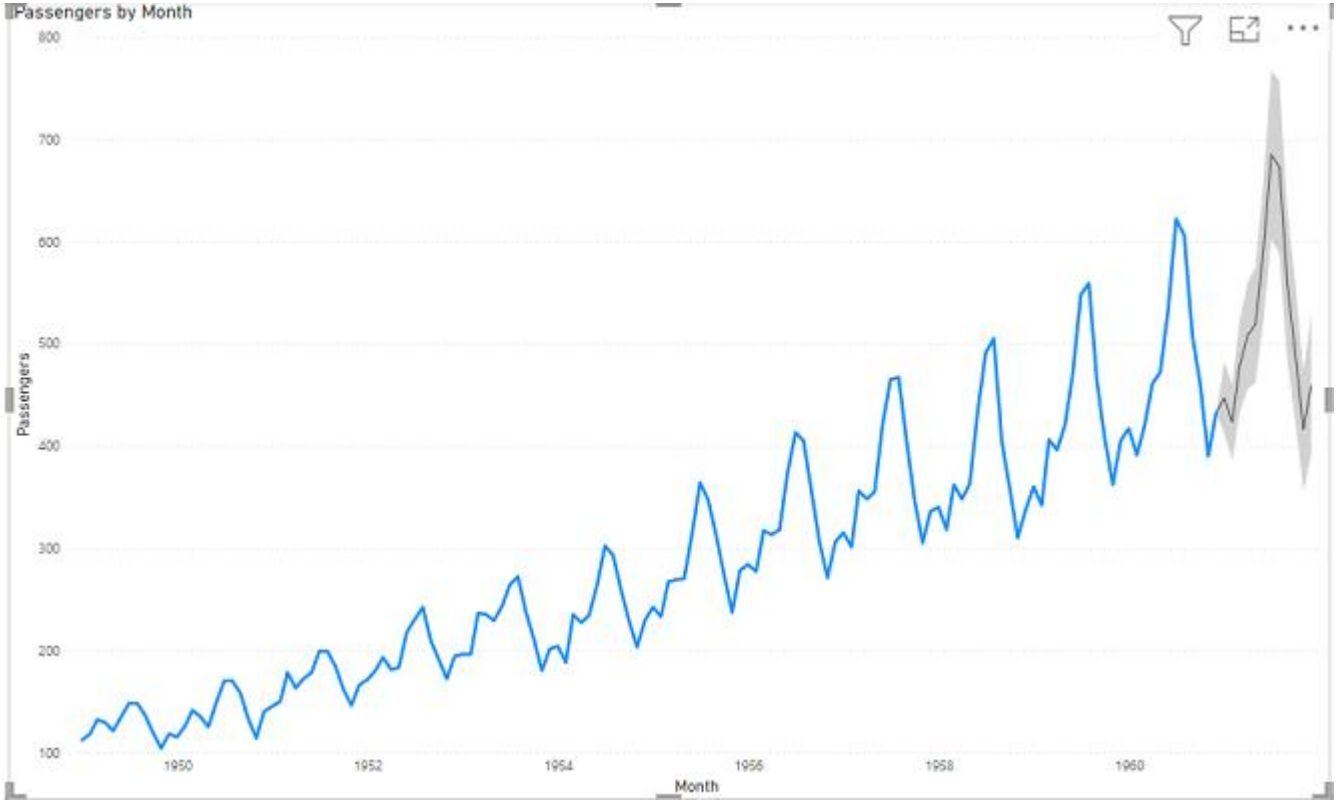




TIME SERIES FORECASTING IN POWER BI

Forecast for 12 points (months) ahead, confidence interval 95%.

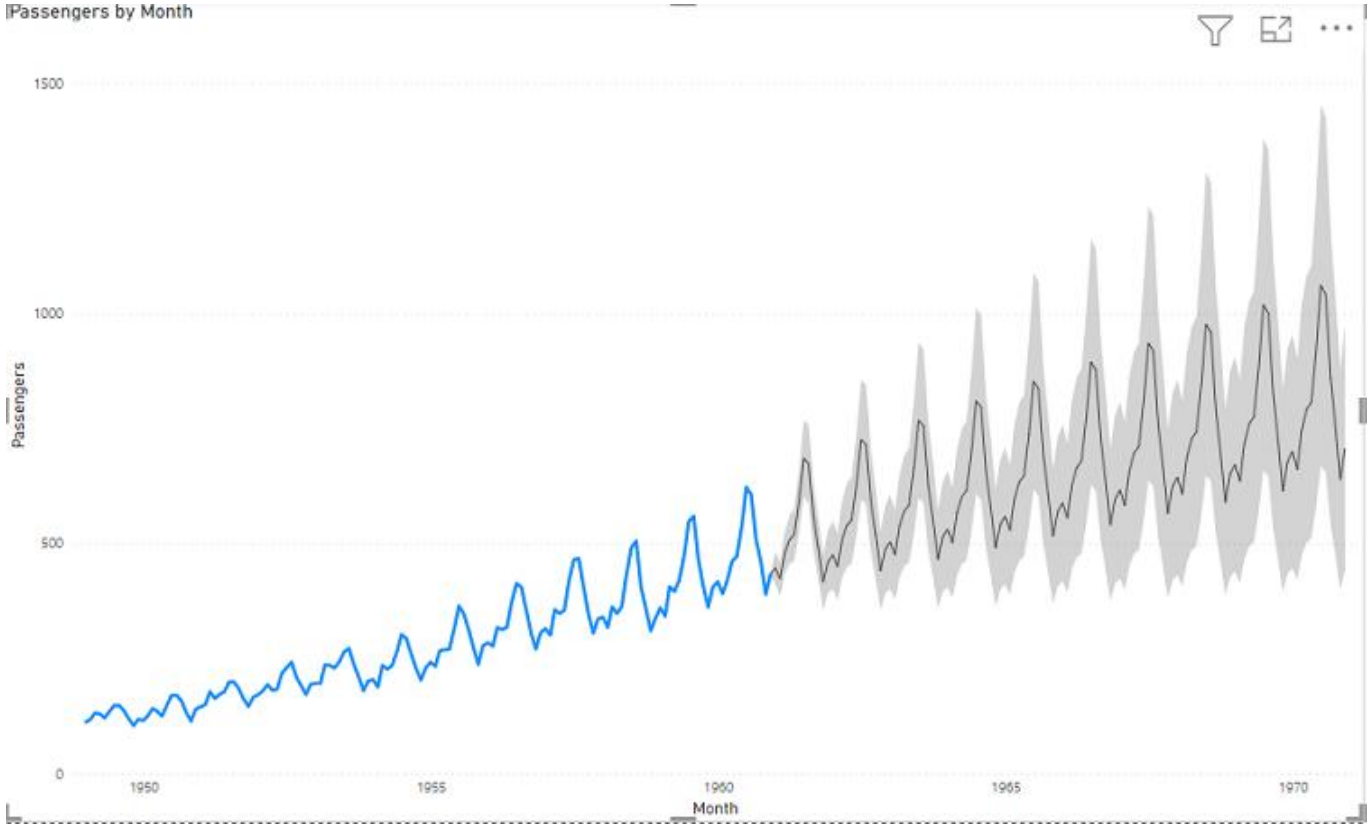
The grey band is where we expect the values to be with 95% confidence.



TIME SERIES FORECASTING IN POWER BI: WHAT IF...

What if we want to forecast very far ahead?

10 years (120 months):

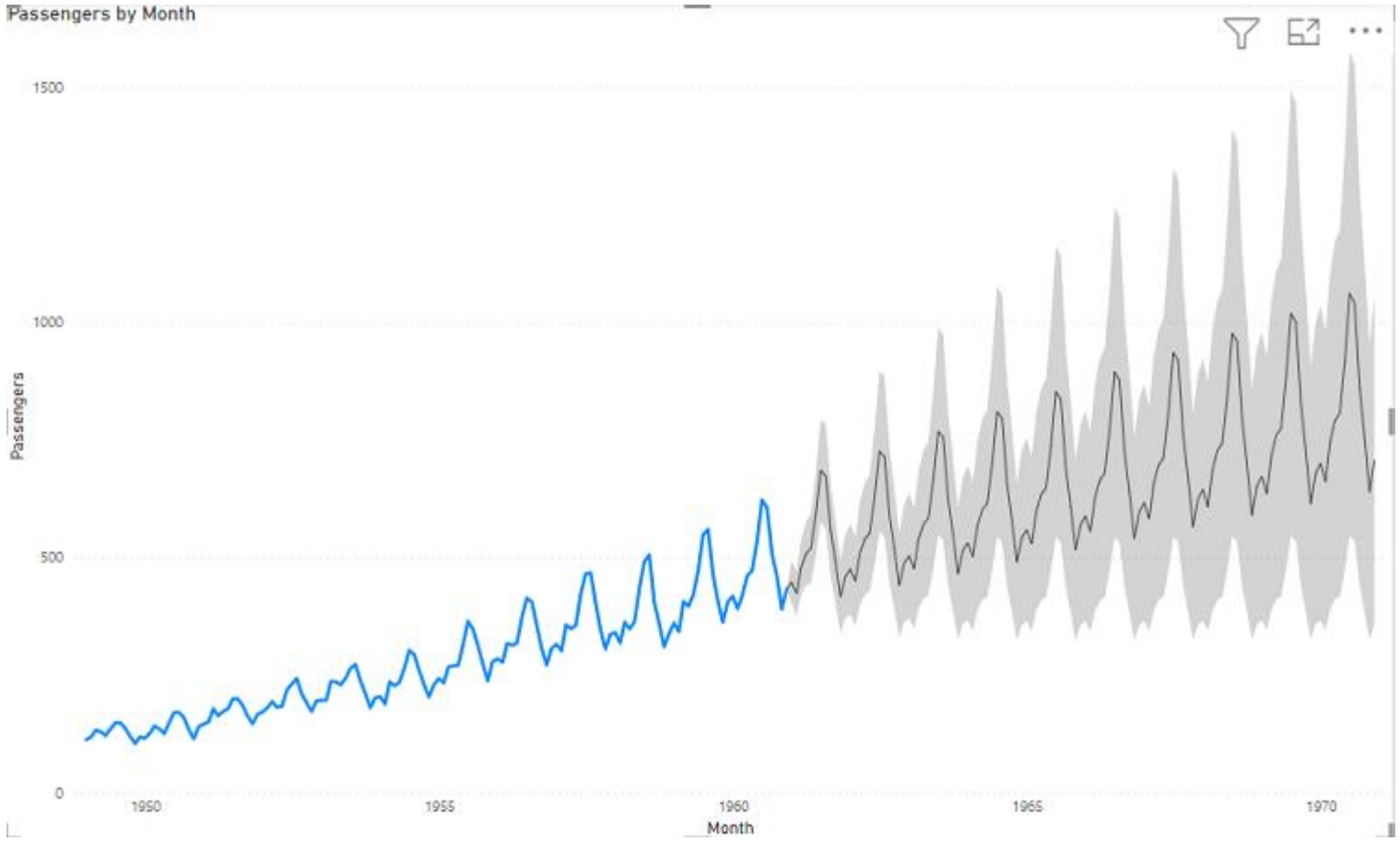


The further ahead we go, the wider the grey band (the 95% confidence interval), the greater the uncertainty of the forecast.

TIME SERIES FORECASTING IN POWER BI: WHAT IF...

What if we want to have maximum confidence?

10 years (120 months) and confidence interval 99%:



The grey band got even wider. The higher the confidence level, the wider the interval.